

EXPECTED VALUE $E(\bar{X})$

"Center of mass" of a probability distribution

$$\Rightarrow E(\bar{X}) = \sum_k x_k P(\bar{X} = x_k) = \mu$$

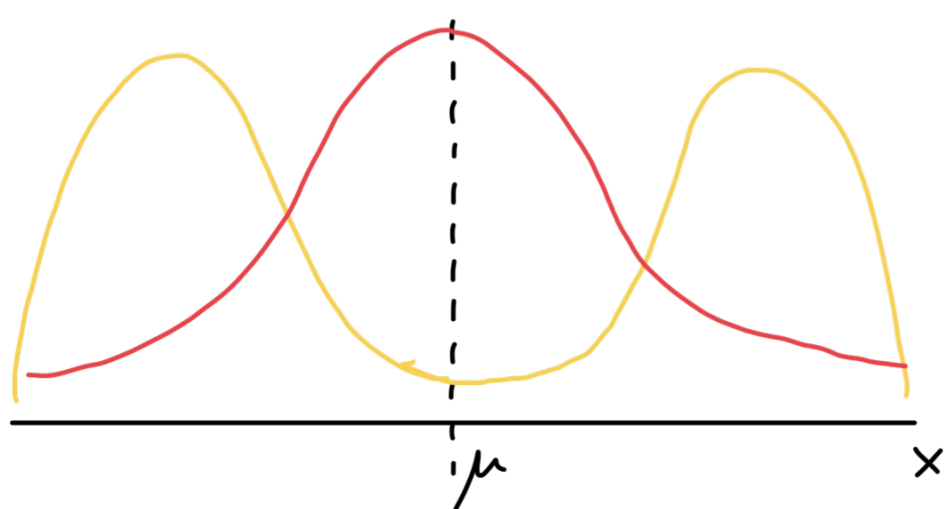
$$\Rightarrow E(\bar{X}) = \int_{-\infty}^{\infty} x f(x) dx$$

If 1 sample $\bar{X}_j$ n times and average

$$\bar{\bar{X}} = \frac{1}{n} \sum_{j=1}^n \bar{X}_j \quad (\text{sample mean})$$

$$\lim_{n \rightarrow \infty} \bar{\bar{X}} \rightarrow \mu = E(\bar{X})$$

ISSUES WITH E.V.



$\Rightarrow$ BOTH DISTRIBUTIONS HAVE SAME μ

PROPERTIES OF THE EXPECTED VALUE

$$① \quad E(\bar{X} + \bar{Y}) = E(\bar{X}) + E(\bar{Y})$$

$$② \quad \text{If } \bar{X}, \bar{Y} \text{ independent } E(\bar{X}\bar{Y}) = E(\bar{X})E(\bar{Y})$$

$$③ \quad \bar{Y} = g(\bar{X})$$

$$E(\bar{Y}) = E(g(\bar{X})) = \sum_x g(x) P(x) \text{ discrete } \bar{X}$$

$$= \int_{-\infty}^{\infty} g(x) f(x) dx$$

$$\underline{E_x}: E(\bar{X}^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$E(\bar{X}\bar{Y}) = \sum_x \sum_y xy \underbrace{P(\bar{X}=x, \bar{Y}=y)}_{\text{if independent}}$$

$$= \sum_x \sum_y xy P(\bar{X}=x) P(\bar{Y}=y)$$

$$= \underbrace{\sum_x x P(\bar{X}=x)}_{E(\bar{X})} \underbrace{\sum_y y P(\bar{Y}=y)}_{E(\bar{Y})} =$$

$$= E(\bar{X}) E(\bar{Y})$$